

It’s the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic

Anonymous submission

Abstract

Vision–language models (VLMs) are the default critic inside slide- and document-generation agents, yet they routinely pass slides whose text visibly overflows its box. We ask whether this is a failure of *perception* (the model cannot see the defect) or of *elicitation* (it sees, but the way we ask buries the signal). We settle this per defect with an attribution protocol built on a lossless structured-geometry oracle, and find the answer is not uniform: some failures are genuinely sub-perceptual, and no prompt recovers them; others vanish the moment we change how we ask, with the model and image frozen. The diagnosis prescribes the critic by routing each defect to the engine that owns its bottleneck, and a minimal symbolic–neural hybrid follows as a consequence rather than a design. For much of what VLMs miss on a slide, it is the elicitation, not the eyes.

1 Introduction

An automatic critic that flags layout and content defects is now a load-bearing component of every slide- and document-generation agent, and the default critic is a general-purpose VLM asked to grade a rendered slide. These critics fail in a revealing way: shown a slide whose title visibly spills out of its box, a strong VLM under a standard rubric prompt answers “no defect.” Whether this is a failure of *perception* (it cannot see the overflow) or of *elicitation* (it sees, but the way we ask buries the signal) is rarely asked, yet the answer decides whether the remedy is a larger model, a different prompt, or a different tool entirely. We treat that question as the design problem and answer it empirically.

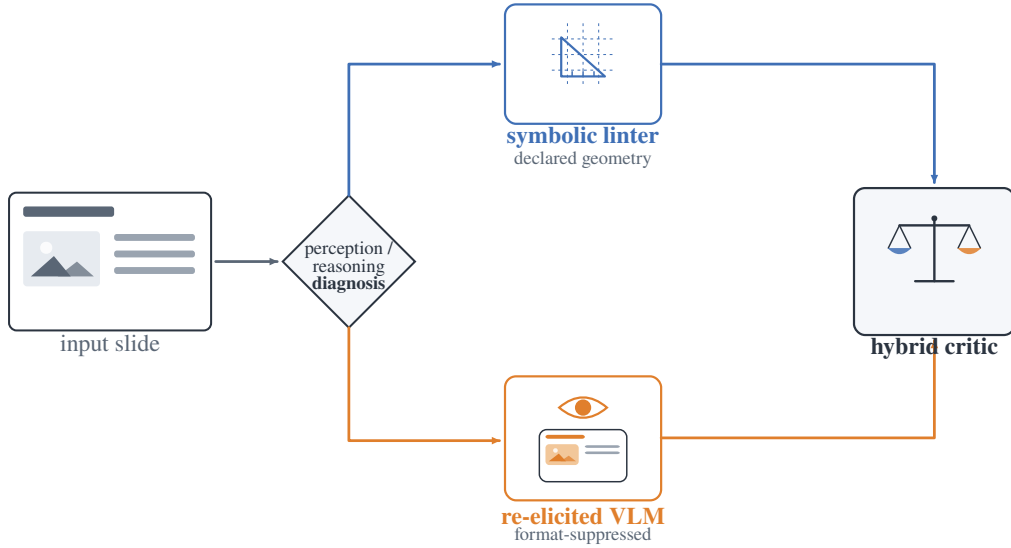
A VLM that “cannot see” a slide defect is hiding two failures of opposite character, and separating them is the whole problem. Some defects are *genuinely sub-perceptual*: a three-pixel alignment offset or a one-point font mismatch sits below the model’s effective acuity, and no amount of prompting, scale, or resolution recovers it. Others are *format-suppressed*: the model can perceive them, but the conventional elicitation (“here is a slide and a rubric; score every defect type in one pointwise pass”) buries the signal, while merely *changing how we ask*, through a forced choice between two candidates or an atomic yes/no per defect type with a required piece of evidence, recovers it with the model and image frozen. Conflating the two is why the field oscillates between “VLMs are bad at layout” and “VLMs can judge slides”; both hold, for different defects.

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We make the split measurable with an attribution protocol built on a *lossless structured-geometry oracle*. Each slide carries an intermediate representation (IR) of element bounding boxes, text, and style; we present the same paired (defective, clean) slides under four input modalities (image-only, structured-oracle, VLM-caption, and image+oracle) crossed with detect/localize/repair tasks. Because the oracle is lossless machine-readable structure rather than a model-written caption, the perception channel it isolates is uncontaminated: a model that fails image-only but succeeds with the structured oracle is bottlenecked on perception, not reasoning. This is the methodological difference from concurrent perception/reasoning attribution that uses natural-language perception oracles, which re-inject the very perceptual error they try to factor out.

The diagnosis then prescribes the critic (Fig. 1): route each defect to the engine that owns its bottleneck. Sub-perceptual geometry goes to a symbolic linter that reads coordinates and rules; text and structural semantics go to a language model; and the format-suppressed, perceivable classes go to a VLM, but only under the changed elicitation the diagnosis shows recovers them. To stress-test the design we construct a render-containment overflow class that is invisible to the linter by construction (the declared box is legal; only the rendered pixels overflow) and that some specialized reward scorers miss while general perceptual scorers and a frontier VLM judge catch, so its detection tracks rendered-quality read-out rather than the training-domain label (Sec. 5.3). The full symbolic–neural critic is built for *IR-owning generation agents*, where the declared structure the linter relies on is available; for bare third-party pixels the deployable critic is the re-elicited VLM alone (Sec. 7), and coverage claims should be read within that scope.

Contributions. Our contributions form a chain. (i) A perception/reasoning attribution protocol built on a *lossless* structured-geometry oracle (element boxes, text, style—not a caption that re-injects perceptual error) that separates VLM inspection failures into *genuinely sub-perceptual* and *format-suppressed*, reproducing on real CC-licensed layouts with no annotation (Secs. 4, 7). (ii) The central finding, *elicitation, not eyes*: for much of what a VLM appears unable to see, changing only how we ask, model and image frozen, recovers detection the pointwise rubric buries—pinned on a con-



G7 render-overflow (legal box, overflowing pixels) is linter-blind by construction; only a rendered-quality read-out catches it

Figure 1: **The paper in one picture.** A perception/reasoning diagnosis (Sec. 4) shows slide-inspection failures are of two kinds. Declared-geometry defects are exact and cheap for a symbolic linter, with a sub-threshold acuity tail nothing recovers; merely format-suppressed defects are recovered by a vision–language model under a changed elicitation. Routing each defect to its bottleneck-appropriate engine yields a hybrid critic (Sec. 5). A constructed render-overflow class (legal declared box, overflowing pixels) is the crux: invisible to the linter by construction, it is caught only by engines with a rendered-quality read-out (a re-elicited VLM, a frontier VLM judge, or general perceptual reward models), not by domain-labelled scorers, so its detection tracks perceptual capability rather than training domain (Sec. 5.3).

structed render class and bounded by where it fails (Sec. 5). (iii) The diagnosis *prescribes* the critic: routing each defect to the engine owning its bottleneck (a linter for declared geometry, one re-elicited VLM for the rest) covers nearly the whole taxonomy and matches a pre-registered routed hybrid, while the linter-blind render class makes the blind spot falsifiable and a reward-side audit reproduces the split (Sec. 5). (iv) An 8B examiner finetuned on injected defects surpasses a zero-shot 30B in-distribution and abstains rather than hallucinates geometry, and a perturbation-fidelity audit shows many injected geometry defects never render, a hazard any IR-labeled dataset inherits (Sec. 6).

2 Related work

The VLM perception bottleneck. A growing line of work argues that VLM failures on visually grounded tasks are dominated by perception rather than reasoning, decomposing the bottleneck for chart understanding (Liu et al. 2025b) or showing that the needed visual information is present in the representation but unused (Fu et al. 2025); MMVP (Tong et al. 2024) is the canonical demonstration that VLMs stumble on minimally-different image pairs humans solve trivially, and its sub-perceptual acuity floor is what our title (“not the eyes”) echoes. We share the perception-first stance but ask it of a different task family, *quality inspection of generated*

artifacts, where the perception channel can be isolated exactly with structured geometry rather than approximated by captions, and where the answer feeds a critic architecture rather than a benchmark.

Perception/reasoning attribution. Concurrent work separates perception from reasoning in abstract visual-reasoning benchmarks via natural-language perception oracles (Wang et al. 2025a). We share the goal but differ in kind: our oracle is a *lossless* structured representation rather than a lossy caption, a cleaner attribution boundary because a caption re-injects the very perceptual error it is meant to factor out; our domain of generated documents admits exact injected labels; and we close the loop from diagnosis to a deployed critic. Layout-error-detection work (Heo et al. 2025) introduces distribution-grounded structural-error injection but for *document-layout-analysis predictions*, and notes that existing metrics cannot isolate whether failures stem from recognition or reasoning, the gap our attribution protocol fills.

Relative vs. absolute judgement. Our finding that relative elicitation beats absolute scoring echoes work showing VLM judges can rank but cannot reliably score (Kumar et al. 2026); that result concerns a judge’s *output mode*, orthogonal to our *source of failure*, and we adopt it to motivate the pairwise and atomic-binary heads of our critic. Comparative/relative

elicitation is also known to recover discriminative signal that pointwise scoring suppresses in LLM/VLM judges (Liusie, Manakul, and Gales 2024), though the protocol choice is not bias-free and can itself distort preferences (Tripathi et al. 2025). Our contribution is not that effect but its *defect-class boundary*: it rescues text overflow (G1) and image/text contradiction (S6) yet not fine alignment (G3) or terminology drift (S3), which is what turns the elicitation effect into a perception/format attribution signal.

Slide generation, evaluation, and reward. Slide generation with design feedback is active: AutoPresent (Ge et al. 2025) treats it as tool-augmented program generation with self-refinement, aesthetic-reward systems (Pan et al. 2026; Liu et al. 2025a) report that iterative VLM refinement helps gross layout damage but fails on fine aesthetics and reward-hacks, and slide-evaluation studies (Kang, Bao, and Goswami 2025) call for calibrated critic-in-the-loop selection. These *motivate* a routed critic rather than compete: they observe the VLM’s perceptual ceiling and route around it with rules, whereas we first *diagnose* that ceiling and show part of it is elicitation-recoverable. We probe real data with SLIDEAUDIT (Zhang et al. 2025) and audit five published scorers plus a closed frontier VLM-as-judge (Sec. 5.3). Defect injection adapts the paired-data critic recipe to rendered documents at zero annotation cost, and we use it as the semantic engine of the routed critic.

3 Problem setup and defect taxonomy

Task and representation. The critic inspects a single rendered slide, optionally with its declared structure, and must *flag and localize* defects of a fixed set of types; a detection counts only when it fires on a defective slide and stays silent on its matched clean twin. Every slide is an intermediate representation (IR) of typed elements with bounding boxes, text, and style; we inject defects programmatically into the IR and render, yielding exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry* group G1–G6, a *semantic* group S1–S6, and one constructed render class G7; it is aligned to the expert-derived SLIDEAUDIT taxonomy (Zhang et al. 2025) via a bidirectional map (crosswalk in the supplement), and results are reported in its canonical classes. We add G7 because it is the sharpest test of the linter’s blind spot: the declared box is legal (inside the safe margin, no overlap) but the rendered content overflows its container, so the defect exists *only* in pixels. On real agent-generated decks G7 is common, affecting most slides of a Zenodo10K slice and over half of an AutoPresent slice.¹

Attribution modalities and metric. To attribute a failure, we feed each paired sample to the model under four inputs: **A** the rendered image alone; **B** the lossless structured oracle (element boxes, text, style from the source IR); **B’** a VLM-written caption (a lossy control); and **C** both image and oracle. Under each we ask the model to detect, localize, and repair the defect. The attribution reads off the A-vs-B

¹93-deck Zenodo10K slice: 81/93 decks affected (857/3167 legal boxes); 14-deck AutoPresent slice: 8/14 decks (20/50 legal boxes).

Table 1: Defect taxonomy. The geometry group is detected symbolically; the semantic group is the examiner’s domain; G7 is constructed to be linter-blind. “Channel” is the engine the diagnosis assigns (Sec. 4–5).

Group	Class	Description	Channel
Geometry	G1 text overflow	text exceeds its box	VLM+linter
	G2 element overlap	boxes intersect	linter
	G3 alignment offset	small mis-alignment	linter
	G4 font-size inconsistency	off-grid font size	linter
	G5 brand-color violation	off-palette color (ΔE)	linter
	G6 margin violation	element past safe margin	linter
Semantic	S1 title/body mismatch	body contradicts title	VLM
	S2 narrative-order break	deck sections out of order	LLM
	S3 terminology inconsistency	term drift across deck	term linter
	S4 density violation	slide over/under-packed	LLM
	S5 missing logic section	dropped argument step	LLM
	S6 image/text contradiction	figure contradicts caption	VLM
Render	G7 containment overflow	legal box, rendered overflow	VLM (atomic)

contrast: if the model fails on the image alone (A) but succeeds given the oracle (B), the bottleneck is *perception*—it could not see the defect from pixels; if it fails under both, the bottleneck is *reasoning*.² The headline metric is *balanced accuracy on the paired clean controls*: a detection counts only if it fires on the defective slide and stays silent on its matched clean twin, so a model cannot score by flagging everything.

Two linters. A geometry linter detects G1–G6 from coordinates and rules (brand color via CIELAB ΔE_{2000} , not RGB distance), at zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus near-duplicate clustering, image-free. Both are the “detectors of record” the diagnosis routes to.

4 Diagnosis: two kinds of blindness

Pointwise image-only inspection floors fine geometry. Under the conventional setup—one slide in, one rubric score out, no reference slide for comparison (we call this *pointwise image-only*)—G2–G6 sit at chance for 4B and 8B models, and a 30B breaks only G1 overflow. The floor is invariant to the obvious levers: swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a native-resolution NaViT, MoonViT) leaves every model at the random line (differing only in abstain-vs-over-flag bias), and raising resolution 1536→2048 changes nothing. The lever is elicitation and magnitude, not encoder or pixel budget, so the floor is a property of the *pointwise* protocol, not a hard capability limit. Either way the right detector for declared geometry is the symbolic linter: it reads coordinates directly at balanced accuracy 0.75–1.0 with zero false positives, cheaper and exact where the IR is available.

But some “perceptual” failures are elicitation artifacts, not blind spots. The same setting hides a second, opposite

²The oracle’s ground-truth labels are stripped from the model’s view by a regression-tested de-leaking pass, so B tests perception of structure, not label leakage. B’ is a control, not a second oracle: a caption re-injects the perceptual error it is meant to factor out, so B’s advantage is by construction.

phenomenon. For G1 overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50, indistinguishable from genuinely sub-perceptual classes, yet a two-alternative forced choice between the defective and clean slide jumps to 1.00: the defect is perceivable, and the absolute pointwise call fails to elicit it. G3 alignment behaves the same way, but only *above a magnitude threshold*: sweeping the injected offset as an internal contrast (one element shifted from its aligned sibling column), the forced choice saturates to 1.00 by 16 px (0.83% of slide width) while the sub-threshold tail (≤ 8 px) stays at chance even side by side. So fine alignment is not a class-level capability floor: it is format-suppressed and recoverable wherever the offset clears acuity, with a genuinely sub-perceptual residue only in the small-offset tail. The connecting thread is a *magnitude-gated* recoverable-versus-sub-perceptual boundary rather than a clean geometric/semantic split, and S3 terminology (below) remains the one defect no elicitation rescues.

A counterexample: terminology. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. The structured-oracle channel, which feeds the model the full deck text with both variants present, also fails (0.56): with both variants already explicit in the input, this failure cannot be OCR, and instead reflects cross-page term-matching that does not survive the yes/no framing. The image channel is consistent with fine-grained glyph discrimination and the oracle channel fails even with the text in hand, so neither route is reliable. S3 leaves the VLM entirely for a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM’s 0.69, a negative result that *re-routes* the class.

A genuine geometric blind spot: G6 page offset. Reposing a defect as a within-slide contrast does not rescue every class; on one it fails outright. We redefine G6 margin as a *page offset*: the whole content block is shifted toward one edge (every element together, so internal alignment is preserved and the defect cannot be misalignment), leaving one margin crowded against (or running off) the edge and the opposite side empty.³ Across all six VLMs and both pointwise framings the page offset stays at chance (roster-mean C3 0.50), and the pairwise test that separates *data artifacts* (S6, which recovers to perfect once the figure is rendered) from genuine blindness leaves G6 on the floor for *every* model: none can tell a page-offset slide from a balanced one until content is clipped by the page boundary. So unlike G3/G5 (format-suppressed) and S6 (a render artifact), page offset is a *genuine* perceptual blind spot (VLMs do not represent global content positioning), which is why declared geometry routes to the linter.

A template pitfall that matters later. Enterprise “snap-to-master” templates re-fit element boxes to a grid before

³The original S6 render omitted the figure element, so the cross-modal contradiction was literally unseeable (recall 0); on the valid-figure corpus the S6 2-AFC reproduces at 0.75–1.00 across all six VLMs, confirming it is a render artifact, not blindness.

rendering, and we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping, while overflow (content-determined) survives, a trap for any dataset that derives labels from the IR but evaluates on the rendered image, quantified and tied to a reward-model failure in Sec. 5.3.

5 From diagnosis to a symbolic–neural critic

The diagnosis says a single model is the wrong unit: route each defect to the engine that owns its bottleneck. We build that critic, validate the elicitation recovery it relies on across many models, and stress it with the constructed render class G7.

5.1 Elicitation: format suppression is not capability

Four ways to ask. Holding the model and image fixed, we compare four ways of prompting it. C0 is the conventional baseline: one call that scores *every* defect type at once against a rubric. C3 is the contrast: one call per defect type, asking only a yes/no question and requiring the model to point at the offending element. C1 (free-form describe-then-classify) and C2 (a synthetic-twin pairwise, where the clean reference is re-rendered to a grid) are two further variants we report for completeness. The scientific test is C3 vs. C0—same model, same image, same taxonomy, differing only in “ask everything at once” versus “one yes/no per type with forced evidence.” If $C3 \gg C0$, the VLM’s silence under C0 was a format artifact, not a capability gap.

C3 recovers the linter-blind class across the capable subset. We scope the claim to models that can see G7 at all: a model counts as “capable” only if, under C3, it both detects the overflow (balanced accuracy ≥ 0.70 and precision ≥ 0.70 on paired clean controls) and points at the overflowing element. This excludes InternVL-8B and Ovis-9B, which stay near chance, a capability boundary rather than a format effect. On the four capable models (three scales, three vendors), C3 recovers what C0 suppresses: balanced accuracy $0.50\text{--}0.79 \rightarrow 0.93\text{--}1.00$, precision $0.88\text{--}1.00$. The effect is cross-model: pairing each slide across the four models, C3 fixes 233 G7 items that C0 missed against 2 reversals ($\chi^2_1=227, p < 10^{-50}$).⁴ The mechanism is simple: C0’s one-call rubric has no slot for G7 (it is a render class, not in the original taxonomy), so the model has no name to attach to what it sees and abstains; C3 asks the question directly.

It is the format, not the eyes. Is it naming the class that helps, or changing how we ask? Two controls separate this. C0+ is C0 with G7 simply added to the rubric’s defect list—same one-call format, now the model has a name for the class. C0_*named* instead asks about G7 alone, on its own yes/no call. Table 2 shows the split: adding the name to the one-call rubric (C0+) recovers *recall* (234 defectives newly

⁴Per-model $p \leq 1.5 \times 10^{-11}$, all surviving Holm correction over the pre-registered 61-test family; the two reversals are on Gemma4-31B (C3 recall 88/90). Full table in the supplement.

Table 2: Naming vs. format on G7. Adding the class name to the one-call rubric (C0+) recovers recall but wrecks specificity (55% clean over-flag); asking about G7 on its own call (C0_named, C3) recovers both. C0_named matches C3, so a clean reference adds nothing for G7.

elicitation (G7)	clean over-flag	bal-acc
C0 (whole-tax., unnamed)	2.4%	0.50
C0+ (whole-tax., G7 named)	55%	0.58–0.82
C0_named (atomic, no ref.)	2.4%	0.93–1.00
C3 (atomic, +forced loc.)	2.4%	0.93–1.00

flagged, $p=7 \times 10^{-71}$) but wrecks *specificity*—it then over-flags clean slides on 55% of pairs versus 2.4% for C3, so balanced accuracy stalls. Asking about G7 on its own call (C0_named) recovers both, and supplying a clean reference slide adds nothing on top. So for G7 the suppressor is the *one-call format*, not perception: the model saw the overflow all along. The boundary: for G1 and S6, naming alone does *not* help—those need the clean reference slide (pairwise choice $\rightarrow$ 0.97–1.00), a milder *availability-of-reference* effect. A clean-vs-clean control rules out a forced-pick artifact (capable models answer “tie” on 92–100% of clean pairs).

No universal elicitation, so the critic routes. The best elicitation is jointly set by model capability and defect class—there is no one prompt, which is why the critic routes per defect and model: the C2 synthetic-twin pairwise rescues declared G1 on capable models (snapping absorbs the overflow and the contrast exposes it), while C1 free-form rescues weak models but collapses on strong ones (precision $\approx$ 0.50). The format effect is not a synthetic artifact: on human-annotated SLIDEAUDIT (no structure) C3 still beats C0 for every class (e.g. brand color, alignment). Absolute geometry on bare real pixels stays modest outside blatant overlap—real fine geometry needs the linter, hence the structure bare images lack, so on image-only real data the hybrid degrades to VLM-only, a ceiling set by missing structure, not method.

5.2 The minimal hybrid critic covers the most

Complementary coverage the diagnosis predicts. We score each critic by *named attribution*: a detection is correct only if it names the right defect type on the defective slide and stays silent on the matched clean slide. We call a class “covered” when this balanced accuracy exceeds 0.75 (with the 95% Wilson lower bound above 0.65, so the cell is not just lucky). Coverage tracks the diagnosis: declared geometry is covered by the linter, the format-suppressed classes by a C3-elicited VLM. The headline is their *combination*—the minimal hybrid of linter plus one C3-elicited VLM covers 8 of 9 classes, matching the pre-registered routed hybrid and against just 2/9 for the conventional pointwise VLM-C0 (Table 3). The nine classes are the image-level ones; G4/S3 are linter-owned and S2/S5 are deck-scope (Sec. 6). Note that this routing needs no per-class tuning: the rule follows from the diagnosis, not from optimizing over class outcomes, so the coverage is what the diagnosis predicts rather than a tuned architecture.

Table 3: Per-class named-attribution balanced accuracy [95% Wilson]; n = paired positives (G1 40, G2 40, G3 70, G5 40, G6 80, G7 40, S1 18, S4 36, S6 12; precision per cell in the supplement (Code and Data Supplement)). The minimal linter+VLM-C3 hybrid matches routed coverage and slightly exceeds its mean. Small S1/S6 cells are routing diagnostics, not confirmatory (those claims rest on the elicitation decomposition and 2-AFC).

Class	linter	VLM-C0	VLM-C3	linter+C3	routed
G1 overflow	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	1.00 [.91,1]	1.00 [.91,1]
G2 overlap	0.80 [.68,.87]	0.71 [.60,.79]	0.86 [.74,.92]	0.80 [.68,.87]	0.80 [.68,.87]
G3 alignment	0.93 [.85,.96]	0.62 [.55,.68]	0.71 [.63,.77]	0.93 [.85,.96]	0.93 [.85,.96]
G5 color	1.00 [.91,1]	0.50 [.46,.54]	0.68 [.57,.75]	1.00 [.91,1]	1.00 [.91,1]
G6 margin	0.75 [.67,.80]	0.51 [.40,.61]	0.51 [.47,.55]	0.75 [.67,.80]	0.75 [.67,.80]
G7 render-overflow	0.00 [0,.54]	0.50 [.46,.54]	1.00 [.91,1]	1.00 [.91,1]	1.00 [.91,1]
S1 title/body	0.50 [.41,.59]	0.94 [.75,.98]	0.83 [.63,.92]	0.83 [.63,.92]	0.94 [.75,.98]
S4 density	0.50 [.45,.55]	0.51 [.44,.59]	0.93 [.81,.97]	0.93 [.81,.97]	0.72 [.60,.80]
S6 image/text	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]	0.50 [.38,.62]
mean / covered	0.66 / 5/9	0.59 / 2/9	0.72 / 4/9	0.86 / 8/9	0.85 / 8/9

A data-driven routing correction. We initially routed S1 title/body mismatch to the text LLM, but the text-only probe over-flags (0.25, precision 0.09) while the image-bearing VLM names it reliably (0.94, precision 0.90): title/body mismatch needs the *rendered layout*, not just the text. We re-route S1 to the VLM under its open pointwise prompt (C0, 0.94) rather than C3 (0.83), since recall is already saturated and C3 only adds false positives, a concrete case of per-class elicitation routing. The one image-level class the hybrid does not cover is S6 image/text contradiction, at 0.50 single-image for every engine; the pairwise test shows the model *does* perceive it (recall 1.0, specificity $\sim$ 0, over-asserting rather than missing), so this is the availability-of-reference effect, not a blind spot, in contrast to the genuine G6 page offset.

5.3 G7: the linter-blind class that specialised rewards also miss

The construction. G7 makes the linter’s blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text; the overflow lives in the renderer, so the linter and any structure-only model are blind by construction—a self-check confirms zero findings on 90/90 G7 defectives.

What kind of reward sees G7? The rebuttal is “just train a reward model” or “use the frontier VLM as judge.” We audit five published scorers plus one closed frontier VLM-as-judge on the same paired slides (does each score clean above defective?): the document reward DocReward (Liu et al. 2025c) (Qwen2.5-VL-3B), two general-multimodal rewards on distinct backbones Skywork-VL (Wang et al. 2025b) and PickScore (Kirstain et al. 2023), two aesthetic/perceptual heads LAION-Aesthetic (LAION-AI 2022) and CLIP-IQA (Wang, Chan, and Loy 2023) on the same CLIP-L/14, and the Qwen3.7-Max judge. Each penalizes several in-taxonomy defects it can see (Table 4), confirming each is a functioning scorer.

Table 4: Multi-reward / VLM-judge audit on G7: paired preference accuracy $P(r_{\text{clean}} > r_{\text{def}})$ (chance = 0.5, 95% CI; $n=90$ non-VLM, $n=24$ judge). $\checkmark/\times$ marks reliable G7 detection (CI above / spanning 0.5). G5 (a visible within-slide color clash) is shown as a functioning-scorer check; each row also penalizes several in-taxonomy defects (full grid in the Technical Supplement).

Reward scorer	category	G5	G7
DocReward-3B	document	0.72	$\times$ 0.48 [.38, .58]
Skywork-VL-7B	general-mm	0.57	$\checkmark$ 0.79 [.69, .86]
PickScore-v1	general-mm	0.70	$\checkmark$ 0.71 [.61, .80]
LAION-Aesth.	aesthetic	0.78	$\times$ 0.57 [.46, .66]
CLIP-IQA	aesthetic	0.65	$\checkmark$ 0.83 [.74, .90]
Qwen3.7-Max judge	frontier VLM	—	$\checkmark$ 1.00 [.86, 1]

The G7 split: perceptual capability, not the domain label. On G7 the scorers split, but *not* along the domain label (Table 4). The cleanest evidence is a dissociation within one backbone: CLIP-IQA detects G7 while the LAION aesthetic head on the *same* CLIP-L/14 is at chance. Both general-multimodal rewards and the frontier VLM judge see the same overflow; the linter, the LAION head, and the document reward miss it. Two side-notes: naming containment in Skywork’s prompt lifts G7 to 0.87 (reinforcement, not rescue), and on G5 (a visible internal chromatic clash) most scorers react as the perceptual-capability reading predicts—the linter still owns G5 at 1.00 by exactness, not because the clash is imperceptible.

What the split implies. The DocReward miss should not be over-read: it is one 3B model, so its capacity and its training objective are confounded, and a larger document reward might well detect G7. The cleaner claim is the dissociation within one backbone: G7 is visible to a general quality probe (CLIP-IQA) on CLIP-L/14 yet invisible to a specialized aesthetic head on the *same* CLIP—the reward-side analogue of the C0→C3 recovery, where the perception is present but the trained head, like the prompt format, buries it. The upshot for the critic: G7 needs an engine with a rendered-quality signal, whether that signal comes from a prompt (C3), a judge (Qwen3.7-Max), or a reward head (Skywork, CLIP-IQA). Elicitation matters even for the frontier judge: asked in its default whole-taxonomy form it scores G7 at chance (0.50), asked atomically it names and localizes it perfectly (1.00).

Perturbation fidelity: 45% of injected geometry never renders. The reward audit ties to the template result of Sec. 4. Rendering every injected defective both faithfully and snapped-to-master, 45% of injected geometry defects (100% of overlap, alignment, margin; 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to all five scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: identical inputs force identical scores, a dataset-integrity hazard, not a reward-model deficiency. This is model-agnostic by construction; any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not

Table 5: Page-semantic inspection, balanced accuracy [95% Wilson] on paired clean (best modality). The finetuned 8B examiner beats both zero-shot baselines, including the 30B, *in-distribution*. Deck-scope S2/S5 are a negative result. n = paired positives (2-AFC rows = forced-choice pairs); the S6 2-AFC ($n=12$) is descriptive only.

class	ft-8B	zs-8B	zs-30B	n
S1 title/body	1.000 [.95, 1]	0.778 [.69, .85]	0.933 [.87, .95]	36
S4 density	1.000 [.97, 1]	0.529 [.50, .58]	0.774 [.69, .84]	60
G1 overflow (2-AFC)	1.000 [.97, 1]	0.975 [.93, .99]	0.700 [.61, .77]	120
S6 image/text (2-AFC)	1.000 [.76, 1]	1.000 [.76, 1]	0.500 [.25, .75]	12
S2 narrative (deck)	0.500 [†] [.45, .53]	0.646 [.53, .72]	0.549 [.42, .66]	36
S5 missing-logic (deck)	0.500 [.47, .53]	0.500 [.47, .53]	0.567 [.45, .68]	60

[†] degenerate (always-flag): the SFT mix had no clean-*deck* negatives.

assumed from the IR, a prerequisite the field largely skips, and why our elicitation experiments render defectives faithfully.

6 The semantic engine: a specialist examiner

The hybrid’s semantic engine can be a zero-shot model, but a small specialist does better on its own ground. We finetune Qwen3-VL-8B with QLoRA on 2,142 supervised records built from injected defects, mixing three task formats: pointwise semantic scoring, geometry questions answered from the structured oracle (with the model taught to abstain on geometry when given only pixels), and pairwise G1/S6 choices. We evaluate on held-out splits.

An 8B examiner surpasses a zero-shot 30B on *in-distribution* semantics. On in-domain held-out page-semantic inspection, the finetuned 8B reaches balanced accuracy 1.000 on both S1 title/body and S4 density, beating both zero-shot baselines including the 30B (Table 5). The headline is S4 density recall (0.0 → 0.97): an 8B examiner surpasses a 4× larger zero-shot model ($p=4.5 \times 10^{-7}$), and both in-distribution contrasts survive Holm correction over the 61-test family. The advantage is not a prompt-format artifact: re-running the zero-shot baselines at the finetuned model’s own trained format collapses them to 0.50 (the format asymmetry favors the baselines), and the finetuned model still wins at each model’s best format. On held-out *severities* the semantic edge holds (0.89 vs. 0.50–0.59).

It abstains instead of hallucinating geometry. Under image-only inspection, the finetuned model stays at 0.50 on G2–G6 with *zero* false positives: trained with image-only abstention targets, it declines to hallucinate geometry from pixels and leaves those classes to the linter, the designed behavior, and the reason the hybrid’s geometry column comes from the linter rather than the VLM.

Two limits on the reading. Two results bound an overbroad reading. First, *deck-level* semantics (those that need the whole deck, not one slide—S2 narrative order, S5 a missing argument step) fail pointwise: the finetuned model flags every deck on S2 (Table 5; the training mix had no clean-deck

negatives) and abstains on S5. These classes never received the pairwise treatment the diagnosis prescribes, and fixing them needs a retrain we did not run. Second, there is a *sim-to-real gap*: on image-only SLIDEAUDIT the finetuned model’s synthetic S4 strength does not transfer (0.50); only the zero-shot 30B shows real image-only density signal ($p=1.1 \times 10^{-5}$). Scale, not finetuning, carries bare-pixel transfer: the examiner’s edge is tied to its training *distribution*.

7 External validity

A real pre-sales agent deck. On a real 16-slide proposal from a deployed PowerPoint agent, the verifiable defect is 20 unfilled template placeholders (“add a title here”) across 7 slides—a semantic-completeness defect the linter is blind to. The zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the *finetuned* 8B, top of the intrinsic ranking, is among the *worst* here (0/7), and the linter’s apparent recall is spurious geometry noise: on a novel defect the ranking is by general-VLM ability, not specialist in-distribution quality—the examiner learned a distribution, not universal quality. Feeding the flagged placeholders back to the generator for one revision drives the defect from 20 to 0.

The structured attribution holds on real layouts. The body runs the A/B/C attribution on synthetic slides; the cleanest discrimination needs it on *real* geometry with a *real* oracle. We obtain it with no human or model annotation: ZENODO10K (Zheng et al. 2025) ships real CC-licensed .pptx with intact element XML, so `python-pptx` extracts a geometry/text/style oracle *lossless with respect to the declared IR* (the IR→render gap that G7 exploits is exactly what this oracle does *not* capture). From 26 real decks we inject one single-shape defect per slide in *PPTX XML space* (overflow, overlap, alignment, font, margin) and render both the clean and defective one-slide deck with the same renderer (LibreOffice), so the paired pixel diff isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs / 3 families). Two results follow. First, the §4 “snap absorbs 45%” hazard is *template-specific*: real free-form decks absorb only 7% (render-fidelity 0.93), a positive control. Second, *all three* attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are image-sufficient (A 0.70–0.74, structure adds nothing); a margin bleed is a perception bottleneck the oracle rescues for weaker VLMs (A 0.59 → B 0.70); and fine alignment stays near chance in *every* channel for *every* model (A=B=C $\approx$ 0.50; a supra-threshold 44px internal-contrast re-run reproduces this, A/B/C 0.51–0.54, $n=41$). This is *not* a hard capability floor (on clean slides the same defect recovers under targeted elicitation once it clears acuity), but on real cluttered layouts the misaligned element cannot be isolated among many genuine ones and specificity collapses, so for declared geometry the symbolic linter remains the reliable detector regardless.

Open-world: structure recovered from pixels does not restore the linter. A third-party PDF/PNG deck ships no IR, so we ask whether *recovering* element structure from pixels re-arms the linter. Following DocLayNet/PubLayNet (Pfaffmann et al. 2022; Zhong, Tang, and Yepes 2019), PP-

DocLayoutV2 returns region boxes per render (class-agnostic NMS removes cross-class duplicates) that feed the *same* linter. The answer is a pre-registered negative: recovered structure leaves the linter at chance on G1/G3/G4 and rescues only overlap partially (G2 0.54 vs. native-IR 0.67 vs. VLM-only 0.87); the detector *over-segments* ($\approx 2\times$ more boxes, recovery recall at $\text{IoU} \geq 0.5$ only 0.30–0.39), so collisions split and overlap recall falls $0.78 \rightarrow 0.29$. On image-only SLIDEAUDIT the recovered linter clears chance only on G2 (0.55) while the C3-elicited VLM reaches 0.94. On real third-party decks even *native* IR is weakened (no position/font bookkeeping, so G3/G4 rules are at chance and margin/overlap rules *over-fire*, specificity 0.11/0.56), so the VLM-only engine *dominates every coarse class*: the linter’s edge is an in-distribution, clean-native-IR phenomenon, and open-world the deployable critic is VLM-bound.

8 Limitations

No human-inspector reference point is provided; the contribution is the *diagnosis* and routing, not a claim to match human inspectors. Three scope bounds: the real-layout attribution (§7) uses *injected* defects (real geometry/renderer/oracle, not naturally-occurring), so image-only SLIDEAUDIT is the complementary real probe; the reward audit’s document-reward miss is instance-level (a second document reward and an InternLM-family reward were not stood up), so the categorical read rests on the two general-multimodal backbones and the same-backbone CLIP-IQA/LAION dissociation; and the examiner→generation downstream effect is near-null (correlates +0.66 under a strong generator but at sub-1% magnitude, collapsing to 0.00 under an unfloored weak generator whose headroom is *deck coverage*, which a layout critic does not move: $\Delta q = +0.00$ (quality change) vs. a coverage critique’s +0.32). Two caveats, both controlled: the 0.50 → 1.00 recovery could in principle be reduced task difficulty rather than elicitation, but the C0+ and guess-floor controls (§5.1) scope the strong format-suppression claim to G7 and relabel G1/S6 as a milder availability-of-reference effect; and the 8/9 coverage (0.86) is a *native-structure* number, an upper bound for IR-owning agents (image-only real decks degrade to VLM-only, §7).

9 Conclusion

A slide-defect critic should not be one model. Telling apart the two blindnesses (a magnitude-gated acuity floor versus format-suppressed perception) turns “VLMs are bad at layout” into a routing rule: declared geometry to a symbolic linter, the perceivable-but-suppressed classes to a VLM under an elicitation that recovers them. A linter plus one re-elicited VLM covers eight of nine classes, matching a pre-registered routed hybrid (Table 3). A constructed render-overflow class makes the linter’s blind spot falsifiable and splits scorers by capability, not domain. The contribution is the diagnosis; the routed critic is its consequence. *Reproducibility*: code, linters, harness, and reward adapters are in the Code and Data Supplement (4×RTX 3080, vLLM, QLoRA Qwen3-VL-8B); all 61 paired tests carry Holm/BH correction.

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